

Unit 7, Lesson 9: Percent Increase and Decrease

Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.

Learning Target

- ☐ I can find percent increase or percent decrease.

7.9.1 Warm-Up: Interpreting Percentages

1. If exactly 93.6% of people completed a survey, what is the fewest number of people that were surveyed? Consider: How could it be determined if only one person could have completed it? Can partial people complete the survey?
2. Explain your answer using mathematics.

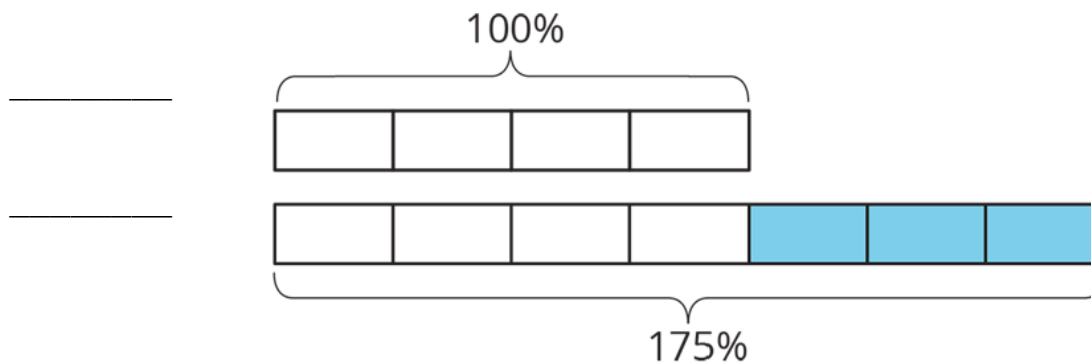
7.9.2 Guided Instruction: Percent of Change

Percentages can be used to make comparisons from one quantity to another.

- It takes Andre $\frac{3}{4}$ times longer than it took Jada to get to school.

The diagram models this in terms of percentages.

- Label each line of the tape diagram with the name of the student who is represented.



- Complete the statements.

The time it takes Andre to get to school is 75%

- ☐ more

☐ less

than the time it takes Jada. The

diagram shows Andre's travel time is

- ☐ 75%

☐ 175%

of Jada's travel time.

- Let x represent Jada's time. Write an expression to model how long it takes Andre to get to school.
- If it takes Jada 8 minutes to get to school, calculate how long it takes Andre to get to school.

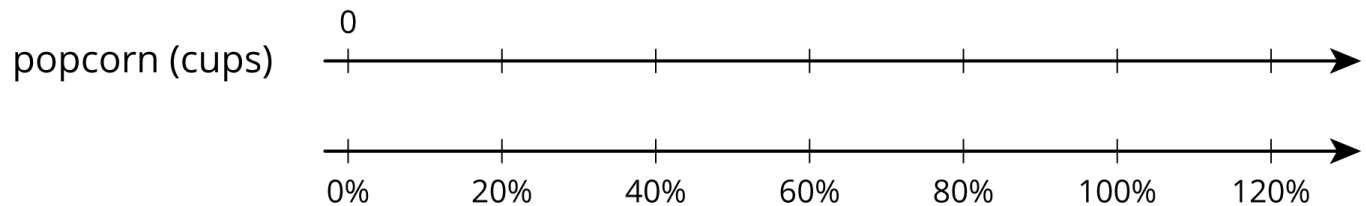
The terms “percent increase” and “percent decrease” are used to describe a specific ***percent of change***.



A ***percent of change*** is the difference between a final value and an initial value, expressed as a percentage of the initial value.

A double number line can also be used to model percent increase or percent decrease.

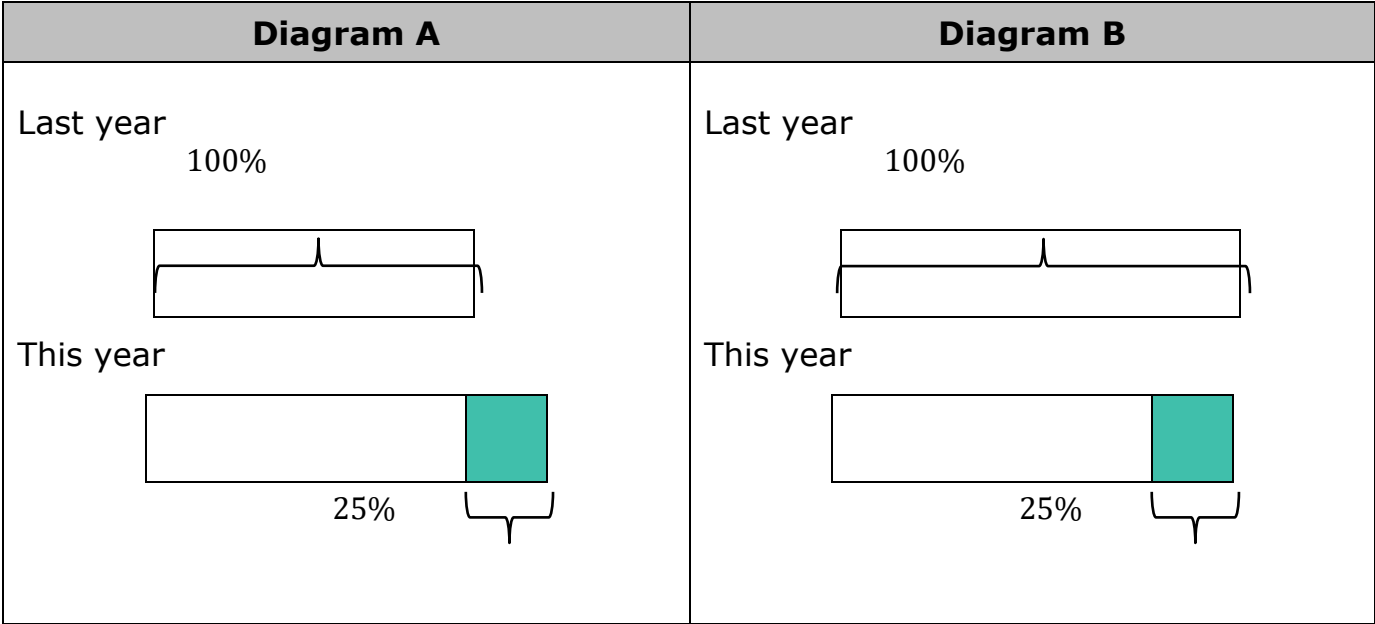
2. A movie theater decreased the size of popcorn bags by 20%. The old bags held 15 cups of popcorn.



- On the top number line, plot the 15 cups of popcorn as the initial value of 100%.
- On the second line, plot 20% less than the initial 100%.
- Use the double number line to find the number of cups in the smaller bag.
- Discuss with your partner how the double number line is helpful for determining the impact of a 20% decrease in the size of the popcorn bag.

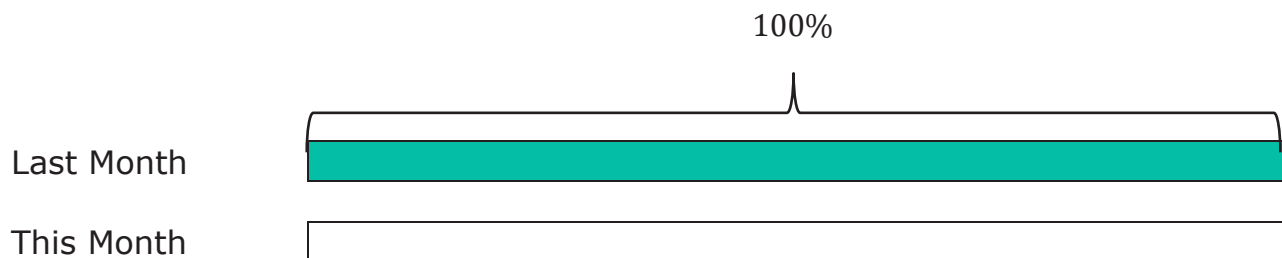
7.9.3 Try It: Percent of Change

1. Explain which diagram, A or B, matches each situation.

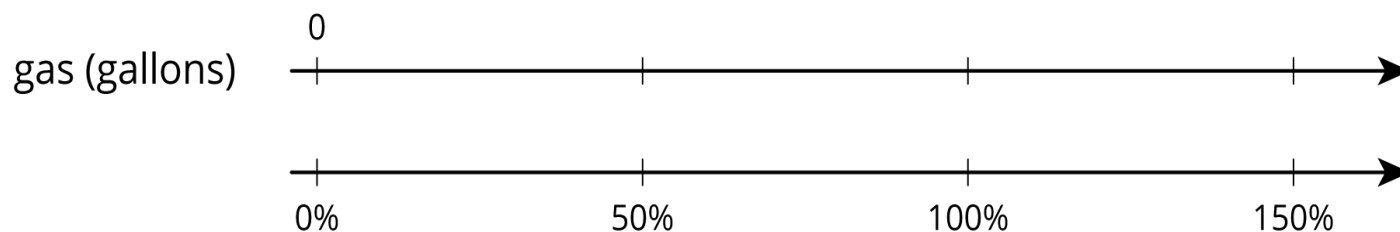


Situation	Matching Diagram	Reason
Compared to last year’s strawberry harvest, this year’s increased 25%.		
This year’s blueberry harvest is 75% of last year’s.		
Compared to last year, this year’s peach harvest decreased by 25%.		
This year’s plum harvest is 125% of last year’s plum harvest.		

2. The number of mosquitoes decreased by 80% from last month to this month. Create a tape diagram of this month to represent the situation.



3. The gas tank in Robert's truck holds 12 gallons. The gas tank in his mom's truck holds 50% more gas. Using the double number lines, determine how many gallons the gas tank of Robert's mom's truck holds.



7.9.4 Collaboration: Calculating Percent Increase or Decrease

Percent increase and decrease can also be found through calculations.

Complete the following with your partner.

1. A new cereal box contains “20% More” cereal. The original cereal boxes contained 500 grams of cereal.
- a. Explain how to determine how much cereal is in the new box.

b. How much cereal comes in the new box?



2. The price of a shirt at ClothesMart is \$18.50, but there is a store coupon that lowers the price by 20%.
- a. Work with your partner to determine how much the shirt is with the coupon.



The table shows two different ways to complete this problem, along with corresponding answers.

Approach 1	Approach 2
The cost of the shirt with the coupon will be 80% of the original cost.	The new cost of the shirt is what’s remaining after the 20% has been deducted from the original cost.
New cost = $0.8(18.50)$	New cost = $18.50 - 0.2(18.50)$

- b. Explain whether or not these equations are equivalent.
- c. Did your answer to Part A match the solution(s) of the equations in the table?



$$\text{Percent of Change} = 100 \times \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}}$$

3. The original cost of pants at ClothesMart is \$30.00, and after a store coupon is applied, the cost is only \$25.50.

- a. Determine the percent of change by filling in the blanks of the percent of change ratio for the situation.

$$\frac{\boxed{} - \boxed{}}{\text{Initial}} = \frac{\boxed{} - \boxed{}}{\boxed{}} = \frac{\boxed{}}{30} = \boxed{}\%$$

- b. Explain whether the percent of change represents a percent increase or decrease based on its value.

- c. Complete the statements.
When calculating the percent of change using the ratio,

a positive value represents a percent

- ☐ increase,
☐ decrease,

and a negative value represents a percent

- ☐ increase.
☐ decrease.

In this context,

it makes sense that the percent change is a percent

- ☐ increase.
☐ decrease.

because the change is the result of a discount.

7.9.5 Your Turn!

1. A school had 1,200 students last year and only 1,080 students this year. What was the percentage decrease in the number of students?
2. A juice box has 20% more juice in its new packaging. The original packaging held 12 fluid ounces. How much juice does the new packaging hold?

7.9.5 Wrap-Up: Error Analysis

1. The number of campers who attend a local summer camp has increased from 1,938 to 2,128. The camp director calculates the percent of increase for an upcoming presentation. Her work is shown.

Explain the camp director's mistake.

$$2,128 - 1,938 = 190$$

$$190 \div 2,128 = 0.089$$

$$0.089 = 8.9\%$$